JAVA PROGRAM REPORT

Multiple Inheritance in Java Using Interfaces

Programming Language:	Java
Topic:	Multiple Inheritance
Program File:	MultipleInheritance.java

Objective

To demonstrate multiple inheritance in Java through interfaces. SmartPhone implements both Camera and MusicPlayer, providing implementations for takePhoto() and playMusic().

Java Source Code

```
interface Camera {
    void takePhoto();
}

interface MusicPlayer {
    void playMusic();
}

class SmartPhone implements Camera, MusicPlayer {

    public void takePhoto() {
        System.out.println("Smartphone takes photos");
    }

    public void playMusic() {
        System.out.println("Smartphone plays music");
    }
}

public class MultipleInheritance {
    public static void main(String[] args) {

        SmartPhone s = new SmartPhone();

        s.takePhoto();
        s.playMusic();
    }
}
```

Program Explanation

1. Camera Interface: Declares takePhoto(), a behavior that implementing classes must define.

2. MusicPlayer Interface: Declares playMusic(), another behavior required from implementing classes.

3. SmartPhone Class: Implements both Camera and MusicPlayer and provides concrete implementations of both methods.

4. Multiple Inheritance: Java classes cannot extend multiple classes, but a class can implement multiple interfaces. This demonstrates multiple inheritance of type/behavior through interfaces.

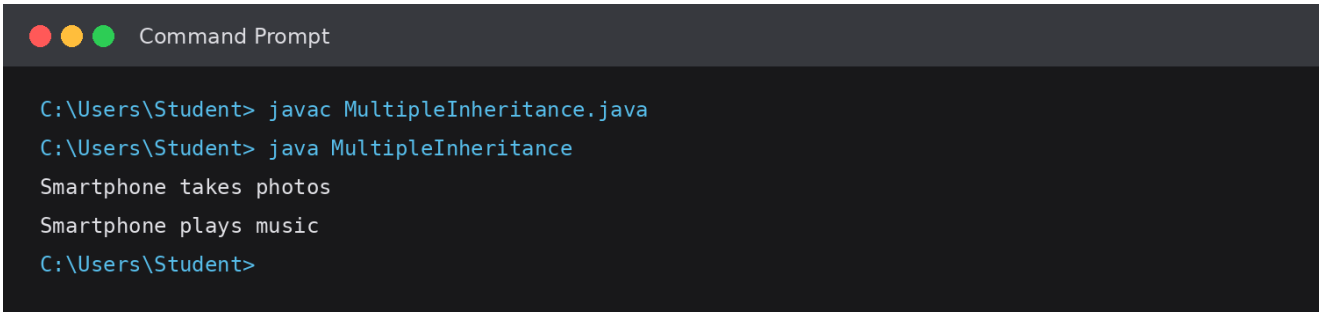
5. Main Method: Creates a SmartPhone object and calls takePhoto() and playMusic().

Expected Output

```
Smartphone takes photos
Smartphone plays music
```

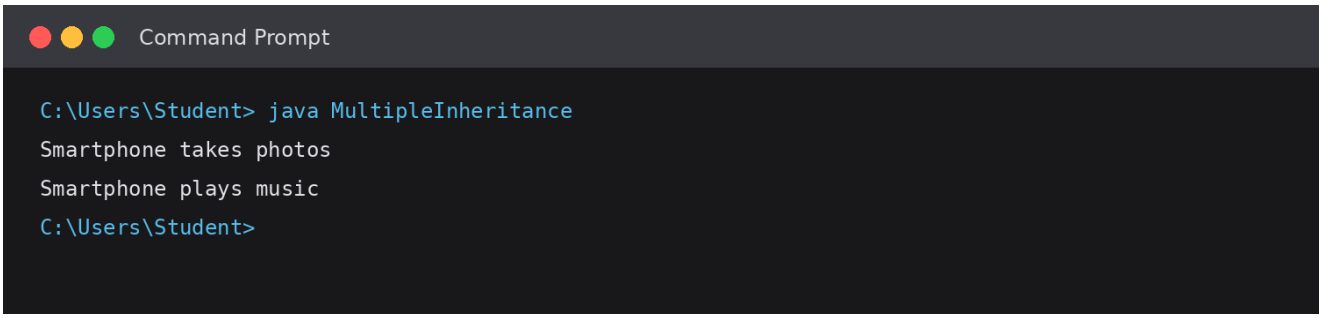
Execution and Output Screenshots

Test Case 1: Compilation and Execution



Verification: Both interface methods execute and display their expected messages.

Test Case 2: Re-execution Verification



Verification: Re-running the class produces the same output sequence.

Test Case Summary

Test Case	Expected Result	Actual Result	Status
Test Case 1	Both interface methods execute.	Output matches expected messages.	**PASS**
Test Case 2	Same output on re-execution.	Output remains consistent.	**PASS**

Conclusion

The program demonstrates multiple inheritance through Java interfaces. SmartPhone implements Camera and MusicPlayer, allowing one class to provide both photo-taking and music-playing functionality. The output confirms that both methods run as expected.